

THESIS DEFENSE

Presented by: Team 5

University of Makati | 2025

OVERVIEW

- INTRODUCTION
- RESULTS AND DISCUSSION
- METHODOLOGY
- SUMMARY OF FINDINGS, CONCLUSION, AND RECOMMENDATION

INTRODUCTION

Traditional identity verification and certification processes in barangay offices often lead to delays and inefficiencies. To address this, our research focuses on developing a web-based barangay system that integrates facial recognition for secure and efficient authentication. By automating identity verification, the system enhances security, reduces processing time, and improves administrative efficiency. This digital transformation ensures a more accessible, reliable, and user-friendly service for both residents and barangay officials.

HYPOTHESIS

- 1.The integration of facial recognition technology in the VerifyID system will enhance the accuracy and efficiency of identity verification within the barangay.
- 2.The automated certificate request management feature will reduce manual processing time and improve administrative efficiency.
- 3.The system's user-friendly design and responsive interface will ensure ease of use and accessibility for both barangay officials and residents.
- 4.The implementation of real-time announcements, news updates, and inquiry handling will improve communication between barangay officials and residents.
- 5.The security features of the system, including SSL encryption and Google Cloud hosting, will effectively protect user data from unauthorized access.

SCOPE AND OBJECTIVE

1. The VerifyID system aims to modernize barangay administrative processes by automating identity authentication and certificate management. It features facial verification for user registration, certificate request processing, real-time announcements, an admin dashboard, and an inquiry feature to openly communicate with the barangay officials.
2. This study focuses on the system's implementation and evaluation in Barangay 781, Zone 85, Sta. Ana, Manila City, assessing its effectiveness in reducing manual verification, enhancing security, and improving efficiency. The research covers development and evaluation from September 2024 to August 2025, with performance measured through user testing and system metrics.
3. The system is developed using HTML5, AJAX v5.3, Laravel v11, Bootstrap v5.3, and Python TensorFlow v3.12 to ensure responsiveness, security, and seamless data handling. GoDaddy provides the domain, while Google Cloud Platform (GCP) handles hosting and SSL encryption. The study is limited to the specified barangay and timeframe, ensuring a focused evaluation of VerifyID's impact on local governance.

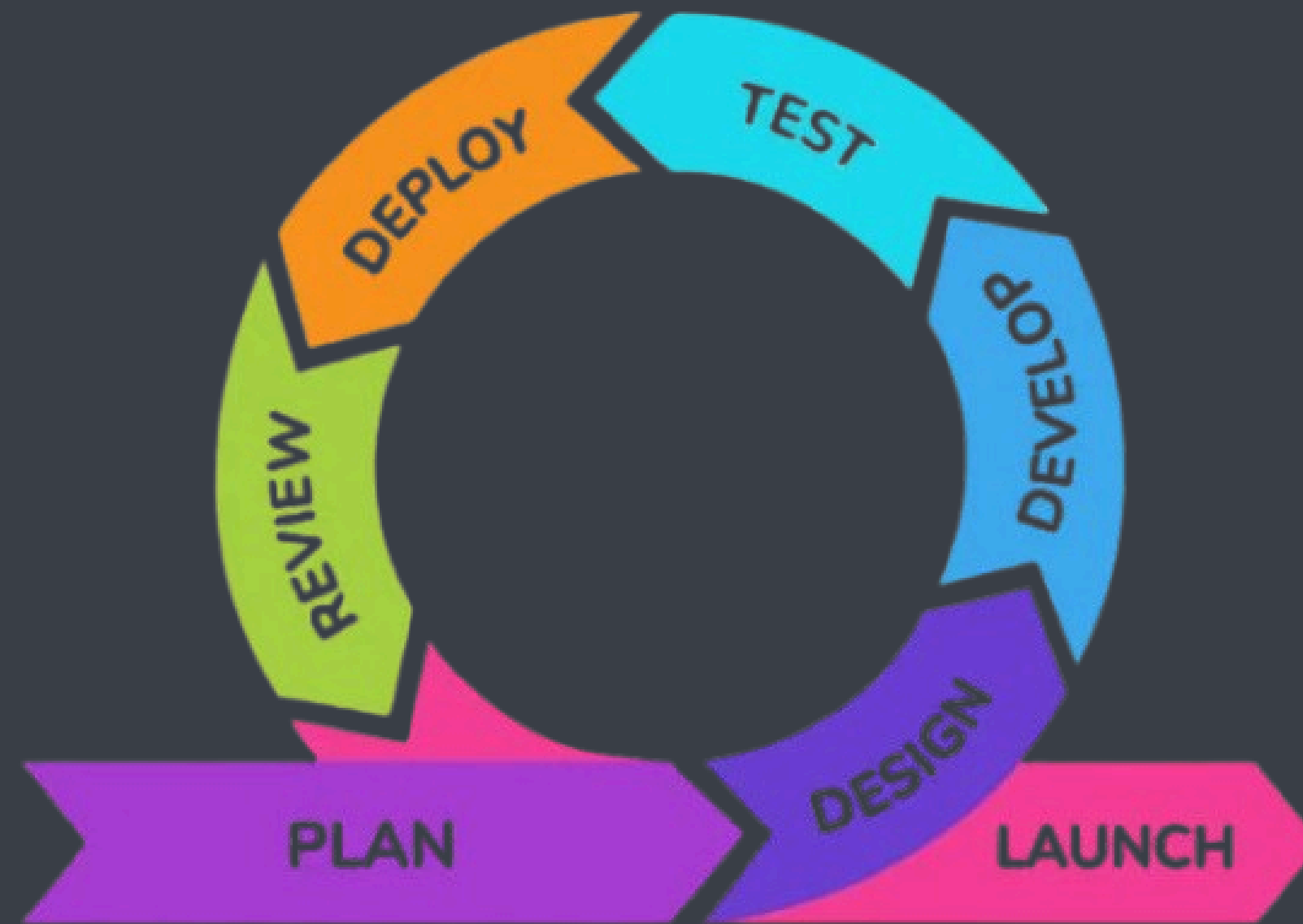
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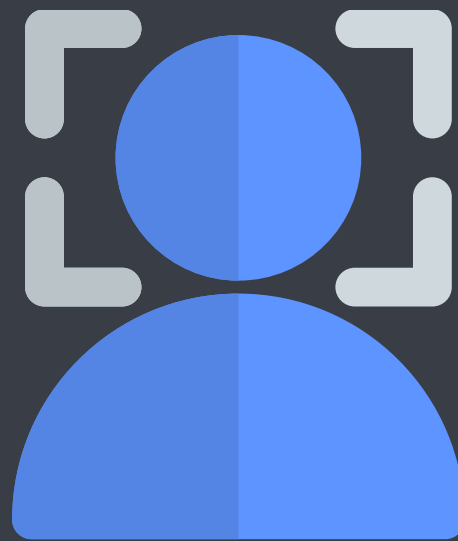
METHODOLOGY



ALGORITHMS



CNN



Faster R-CNN



YOLOv8

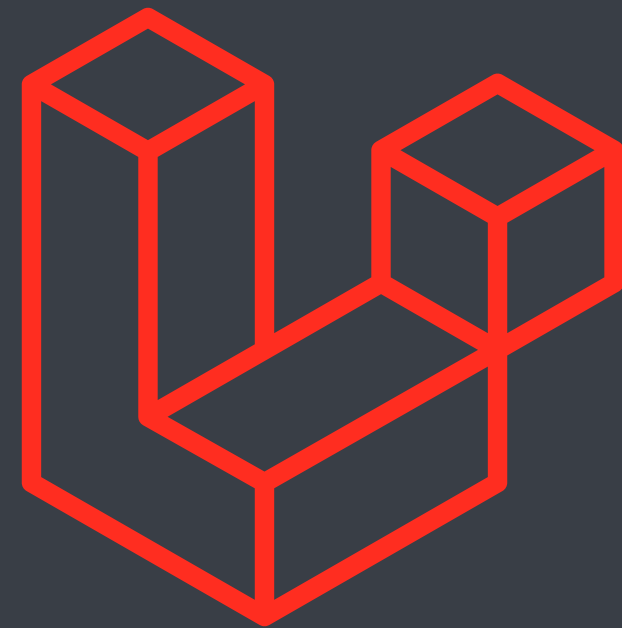
FRAMEWORKS



Bootstrap



TensorFlow

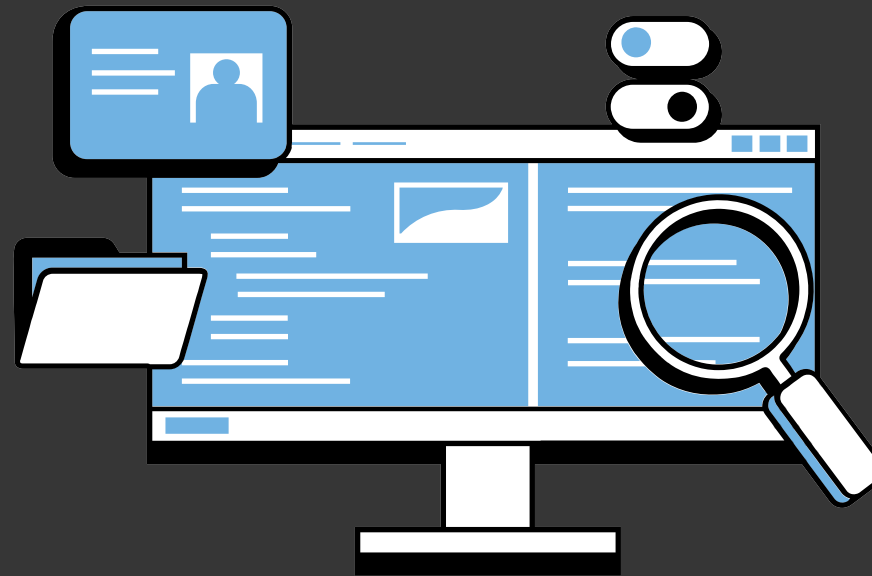


Laravel

RESEARCH DESIGN



Descriptive
Research



Developmental
Research



Experimental
Research

METHODOLOGY



DATA COLLECTION METHOD



Survey



ANALYTICAL TECHNIQUES



Descriptive analysis

MAIN FINDINGS

1

The evaluation of VerifyID confirmed its accuracy, efficiency, and usability in automating identity authentication and barangay services. The CNN model achieved perfect accuracy (1.000), while Fast-R CNN and YOLOv8 showed lower but functional performance.

2

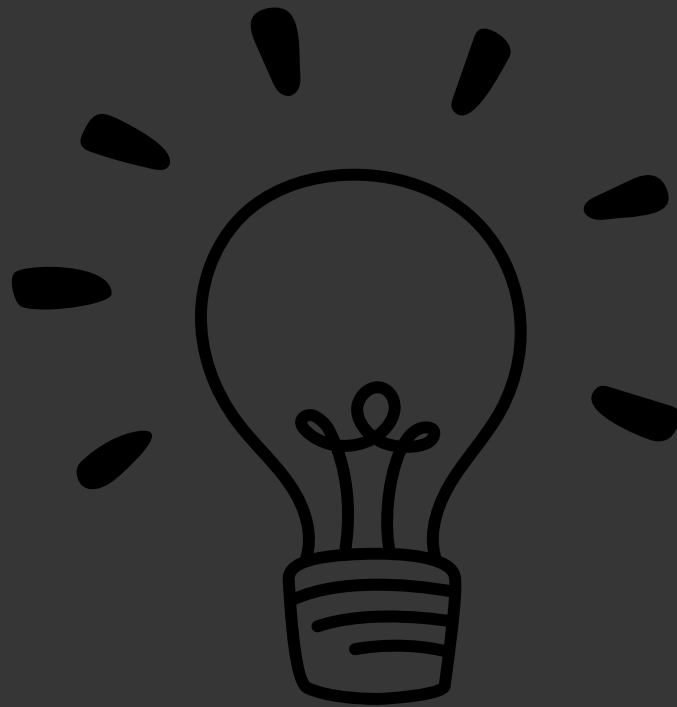
User testing results indicated high satisfaction, with 98.9% confirming accurate verification, 97.8% acknowledging efficient processing, and 96.7% approving multi-platform compatibility. IT experts acknowledges the security but recommended stronger encryption.

SIGNIFICANCE OF THE RESULTS

The study confirms that VerifyID enhances identity authentication, efficiency in barangay processes, and system usability. Results showed 98.9% accuracy in identity verification, 97.8% efficiency in certificate processing, and 96.7% seamless multi-user access. These findings highlight the system's reliability, security, and scalability in local governance. By reducing manual workload, minimizing errors, and improving service accessibility, VerifyID optimizes barangay operations and enhances community engagement. Future improvements, including encryption enhancements and mobile app development, will further strengthen its impact.

CHALLENGES

Poor Lighting



Face Partially Covered



Background noise



SOLUTION

Applied pre-processing (e.g.,
brightness adjustment



Cropped ID for cleaner dataset



Enhanced Face Detection Algorithm

CONCLUSION

The study successfully developed VerifyID, an online barangay system integrating facial recognition for identity authentication and digital certificate management. The system was evaluated based on ISO/IEC 25010 quality metrics, emphasizing functional suitability, efficiency, security, usability, and compatibility. Findings revealed that 98.9% of users confirmed accurate identity verification, while 97.8% found the system effective in streamlining certificate requests. Additionally, 96.7% of respondents approved its multi-platform functionality, ensuring accessibility across devices. The CNN model achieved a perfect accuracy of 1.000, validating its highly reliable facial recognition capabilities. The study concluded that VerifyID enhances barangay service efficiency, reduces manual verification processes, and improves security. While challenges such as accessibility for non-tech-savvy users and security enhancements were identified, proposed future improvements include stronger encryption, real-time feedback mechanisms, and mobile application development to further optimize user experience and system reliability.



THANK YOU

THESIS PRESENTATION

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